CANSAT 2024
REQUIREMENTS

European Space Agency



Introduction

General information about the CanSat project

CanSat is a simulation of a real satellite scaled down to the size of a Matte can. This nano-satellite will be launched at an altitude of 400-600m with rockets of the Space Race Project (Rockets made by new EPFL Rocket Team members). During the descent, it will collect atmospheric measures and mores.

This project aims to let student have a hand on experiment with all the major sub systems and design methods of a satellite such as:

- Structure and CAD
- Power supply
- Telecommunication
- On board computer and sensors
- Flight software
- Ground segment
- Communication and finance

Various workshop will take place during the first semester to guide you through the conception of your satellite. It will also give you a small introduction of the various pole composing the EPFL SpaceCraft Team if you are interested in joining some projects such as CHESS.

EPFL Space Craft Team pole	Pole leader
Management	@Lina Kuhlmann
Systems Engineering	@Alvaro Martinez-Vizmanos
Structure	@Eric
ADCS	@Tanguy Classen
Telecom	@Jeremy Weill
Flight Software	@Raph
OBC	@Oliver Pineda Suárez
EPS	@Lucile Siegfried
Ground Segment	@Adrien Dupont
CanSat	@Alec Horth
Sponsoring	@Angelina Frolova
Communication	@Alix Schmutz

The following requirements are made according to the ESA official requirements , this way the Cansat is compatible with bigger rockets and an opportunity will be given to the best team to fly with the TU Darmstadt rocket team at Euroc 2024 in Portugal.

Structure and recovery

Essential requirements for compatibility with the rockets and your CanSat well-being !!!

- All the components of the CanSat shall fit inside a cylinder of **113-115 mm height** and **64-66 mm diameter.** An additional **35mm** in height is allowed for the retrieval system (ex: parachute)
- The mass of the CanSat shall be between a **minimum of 300 grams** and a maximum **of 350 grams**. CanSat that are lighter shall take additional ballast with them to reach the 300 grams minimum mass limit required.
- The Cansats shall correctly fit inside its assigned rockets. (collaborate with the Space Race teams) and the teams **MUST** discuss about the ejecting mechanism and no component shall be damaged
- The antennas, transducers and other elements of the CanSat shall not extend beyond the can's diameter until it has left the launch vehicle.
- The heat inside the CanSat shall not exceed 50°C (with no ventilation, it can heat up till 80°C…)
- The CanSat shall be dust and moisture resistant
- The CanSat shall be able to withstand an acceleration of up to **15G** and not be damaged when hitting the ground
- The CanSat should have a **recovery system**, such as a parachute. It is recommended to use bright colored fabric, which will facilitate recovery of the CanSat after landing.
- The parachute connection shall be able to withstand at least **40 N** of force. The strength of the parachute shall be tested to ensure that the system will operate nominally
- The Rate of descent of the recovery system shall not exceed **9 m/s**
- The CanSat must have a positional recovery system (bright parachute color, GPS, beeper, …)
- Explosives, detonators, pyrotechnics, and inflammable or dangerous materials are strictly forbidden. All materials used shall be safe for the personnel, the equipment, and the environment.

Feature
Base features
• The data acquisition rate is recommended to be at least 5Hz • The OBC shall have at least 4MB of EEPROM storage • The OBC shall be able to run a real time flight software • The CanSat shall be powered by a battery and/or solar panels. It must be able to run for **at least 4 continuous hours** • The battery shall be accessible from the side or bottom of the CanSat **and easily replaced** • The cansat shall have an accessible power switch • The cansat shall **measure at least one environmental parameter** (temperature, pressure, humidity, …) • The data acquisition shall be done at least during the entirety of the CanSat descent (ascent data acquisition is optional). • Data visualization can be done either live (during operation), post-flight when the data is recovered or both. • Telemetry shall not interfere with the rocket telemetry • A ground station to interact with the CanSat shall be implemented (a ping function is highly recommended to implement in order to check if your CanSat is operational in the rocket) • During Euroc, the rocket will provide a power supply line, so the Cansat can have an external power connector • A communication bus will also be available between the Euroc rocket avionics and the CanSats. That way the rocket telemetry and sensor data will also be available to the Cansats. The interface is to be defined with the TUDsat team Optional Features (some ideas. Not limited to these) : - camera (and camera stabilization) - measure wind speed and direction - telecommunication (telemetry, answer ping from ground station, error correction code, data compression, data encryption, …) - visualization of the estimated position of the cansat in real time - other environmental data retrieval

Logistic

Refer to the CanSat planning for major deadlines

- Cost: **250 CHF** per CanSat from EST. The maximal cost shall not exceed 750 CHF in case the team wants to add their own resources. (The teams should make a list of the components needed and explain their functionality for the PDR. It will be reviewed by EST management before ordering if all is in order.)
- The CanSat shall be flight-ready upon arrival at the launch campaign.
- The CanSat shall be entirely reusable for multiple launches the same day.
- Launch height will be between 400m and 600m for the flight with the space race and around 3000m during Euroc.
- During Euroc, the Cansat will not be deployed because of the high altitude, it will remain inside the rocket.
- The Cansat teams shall assume that it is their responsibility to manage travel and accommodation to the different launch sites.